C

ACT SUBJECT TEST — MATH 2 • UNIT 1.01

# Complex Numbers & De Moivre Theorem

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

**Total Questions:** 25 • Verified Step-by-Step Solutions

## QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

| | | | | |
|---|---|---|---|---|
| Q1 A | Q2 A | Q3 A | Q4 A | Q5 A |
| Q6 A | Q7 A | Q8 A | Q9 D | Q10 C |
| Q11 D | Q12 A | Q13 C | Q14 B | Q15 C |
| Q16 C | Q17 A | Q18 C | Q19 C | Q20 C |
| Q21 B | Q22 D | Q23 B | Q24 C | Q25 C |

## Complex Numbers & De Moivre Theorem Practice Exercises

**1** MCQ

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

- **A** $3 + 4i$ ✓
- **B** $3 - 10i$
- **C** $7 + 4i$
- **D** $3 - 4i$

**WORKED SOLUTION & VERIFICATION**

To subtract complex numbers, distribute the negative sign across the second term and combine like terms (real parts with real parts, and imaginary parts with imaginary parts):

$(5 - 3i) - (2 - 7i) = 5 - 3i - 2 + 7i$

Group the real and imaginary components:

- Real part: $5 - 2 = 3$
- Imaginary part: $-3i + 7i = 4i$

Combining these gives $3 + 4i$.

Common misconception checks:

- Choice B $(3 - 10i)$ occurs if the negative sign is not distributed to the imaginary term, mistakenly calculating $-3i - 7i = -10i$.
- Choice C $(7 + 4i)$ results from accidentally adding the real parts $(5 + 2 = 7)$ while subtracting the imaginary parts.
- Choice D $(3 - 4i)$ results from a sign error when computing $-3 - (-7)$.

**2** MCQ

For what real value of $k$ is the complex expression

$$\frac{k+8i}{3-i}+\frac{2-ki}{3+i}$$

a real number?

- **A** 11 ✓
- **B** 1
- **C** 6
- **D** -11

**WORKED SOLUTION & VERIFICATION**

To determine when the sum of the two fractions is a real number, simplify the expression by combining terms over a common denominator or multiplying by conjugates.

The common denominator of the two fractions is:

$$(3-i)(3+i)=3^2-i^2=9-(-1)=10$$

Rewrite each term over this denominator:

1. For the first term:

$$\frac{(k+8i)(3+i)}{(3-i)(3+i)}=\frac{3k+ki+24i+8i^2}{10}=\frac{(3k-8)+(k+24)i}{10}$$

2. For the second term:

$$\frac{(2-ki)(3-i)}{(3+i)(3-i)}=\frac{6-2i-3ki+ki^2}{10}=\frac{(6-k)+(-3k-2)i}{10}$$

Now, sum the real and imaginary components:

$$\text{Real part}=\frac{(3k-8)+(6-k)}{10}=\frac{2k-2}{10}$$

$$\text{Imaginary part}=\frac{(k+24)+(-3k-2)}{10}=\frac{-2k+22}{10}$$

For the resulting complex number to be a real number, its imaginary part must equal $0$:

$$\frac{-2k+22}{10}=0 \implies -2k+22=0 \implies 2k=22 \implies k=11$$

*Common Misconceptions:*

- Choosing **1** results from setting the real part equal to $0$ $(2k-2=0)$, which would make the expression purely imaginary rather than real.
- Choosing **6** results from equating the real and imaginary parts $(2k-2=-2k+22)$.
- Choosing **-11** results from an algebraic sign error when solving $-2k=-22$.

**3** MCQ

Let $z$ be a complex number satisfying the equation

$$|z - 12| + |z + 12| = 26.$$

What is the maximum possible value of $|z - 5i|$?

- **A** $\frac{169}{12}$ ✓
- **B** $\sqrt{194}$
- **C** 18
- **D** 10

**WORKED SOLUTION & VERIFICATION**

1. **Identify the geometric locus:**

The equation $|z - 12| + |z + 12| = 26$ represents an ellipse in the complex plane with foci at $z_1 = 12$ and $z_2 = -12$ along the real axis.

- The focal distance is $2c = 24 \implies c = 12.$
- The major axis length is $2a = 26 \implies a = 13.$
- The semi-minor axis length is $b = \sqrt{a^2 - c^2} = \sqrt{13^2 - 12^2} = \sqrt{25} = 5.$

In Cartesian coordinates $z = x + iy$, the equation of this ellipse is:

$$\frac{x^2}{13^2} + \frac{y^2}{5^2} = 1, \quad \text{with } -13 \le x \le 13 \text{ and } -5 \le y \le 5.$$

2. **Express the distance function:**

We want to maximize the distance from $z$ to the point $5i = (0, 5)$:

$$D^2 = |z - 5i|^2 = x^2 + (y - 5)^2.$$

From the ellipse equation, $x^2 = 169\,(1 - \frac{y^2}{25}) = 169 - \frac{169}{25}y^2.$

Substituting $x^2$ into $D^2$:

$$D^2(y) = 169 - \frac{169}{25}y^2 + y^2 - 10y + 25 = -\frac{144}{25}y^2 - 10y + 194.$$

3. **Optimize the quadratic over $[-5, 5]$:**

Since the coefficient of $y^2$ is negative, $D^2(y)$ is a downward-opening parabola.

The vertex occurs at:

$$y = -\frac{-10}{2\,(-\frac{144}{25})} = -\frac{10 \times 25}{288} = -\frac{125}{144}.$$

Because $-5 \le -\frac{125}{144} \le 5$, this critical point lies within the domain.

4. **Calculate the maximum value:**

$$D^2\,(-\frac{125}{144}) = -\frac{144}{25}\,(-\frac{125}{144})^2 - 10\,(-\frac{125}{144}) + 194$$

$$= -\frac{625}{144} + \frac{1250}{144} + 194 = \frac{625}{144} + 194 = \frac{625 + 27936}{144} = \frac{28561}{144}.$$

Taking the square root:

$$D = \sqrt{\frac{28561}{144}} = \frac{\sqrt{13^4}}{12} = \frac{169}{12}.$$

**Common Misconceptions:**

- **Choice B ($\sqrt{194}$):** Evaluates the distance at the major axis vertices $z = \pm 13$, giving $\sqrt{13^2 + 5^2} = \sqrt{194} \approx 13.93,$

failing to find the true off-axis global maximum $\frac{169}{12} \approx 14.08$.
- **Choice C ($18$):** Naively applies the triangle inequality $|z - 5i| \leq |z| + 5 \leq 13 + 5 = 18$, which cannot be achieved because the phases of $z$ and $-5i$ cannot align on the ellipse.
- **Choice D ($10$):** Evaluates at the opposite minor axis vertex $z = -5i$, yielding $|-5i - 5i| = 10$.

**4** MCQ

For the imaginary unit $i$ such that $i^2 = -1$, what is the product $(4 - 3i)(2 + 5i)$?

| | |
|---|---|
| **A** $23 + 14i$ ✓ | **B** $-7 + 14i$ |
| **C** $23 - 14i$ | **D** $8 - 15i$ |

**WORKED SOLUTION & VERIFICATION**

To find the product of the two complex numbers, expand using the distributive property (FOIL):

$$(4 - 3i)(2 + 5i) = 4(2) + 4(5i) - 3i(2) - 3i(5i)$$

Multiply the individual terms:

$$= 8 + 20i - 6i - 15i^2$$

Substitute $i^2 = -1$:

$$= 8 + 14i - 15(-1) = 8 + 14i + 15 = 23 + 14i$$

Common errors:
- $-7 + 14i$ results from forgetting that $i^2 = -1$ changes the sign of $-15i^2$, incorrectly computing $8 - 15 = -7$.
- $8 - 15i$ results from only multiplying the first terms and last terms, omitting the cross terms.

5 MCQ

For the imaginary unit $i$, where $i^2 = -1$, the expression $\frac{k+6i}{3-2i}$ is equal to a real number for some real constant $k$. What is the value of $k$?

A -9 ✓

B -4

C 4

D 9

**WORKED SOLUTION & VERIFICATION**

To simplify the quotient of two complex numbers, multiply the numerator and denominator by the complex conjugate of the denominator, $3 + 2i$:

$$\frac{k+6i}{3-2i} = \frac{(k+6i)(3+2i)}{(3-2i)(3+2i)}$$

Expand the denominator:

$$(3-2i)(3+2i) = 3^2 - (2i)^2 = 9 - 4(-1) = 9 + 4 = 13$$

Expand the numerator:

$$(k+6i)(3+2i) = 3k + 2ki + 18i + 12i^2 = 3k + 12(-1) + (2k+18)i = (3k-12) + (2k+18)i$$

Thus, the expression can be written in standard form $a + bi$ as:

$$\frac{3k-12}{13} + \frac{2k+18}{13}i$$

For this expression to be a real number, its imaginary part must be zero:

$$\frac{2k+18}{13} = 0 \implies 2k + 18 = 0 \implies 2k = -18 \implies k = -9$$

Common Misconceptions:

- Setting the real part to zero (confusing "real number" with "purely imaginary") gives $3k - 12 = 0 \implies k = 4$.
- Making a sign error when multiplying by the conjugate leads to $-2k + 18 = 0 \implies k = 9$ or $3k + 12 = 0 \implies k = -4$.

**6** MCQ

In the complex plane, the complex numbers $z$ and $w$ satisfy $|z| = 5$, $|w| = 7$, and $|z + w| = 10$. What is the value of $|z - w|$?

- **A** $4\sqrt{3}$ ✓
- **B** $2\sqrt{6}$
- **C** $4\sqrt{6}$
- **D** $2\sqrt{37}$

**WORKED SOLUTION & VERIFICATION**

To find $|z - w|$, we use the algebraic properties of the modulus and complex conjugates, which correspond to the parallelogram law in the complex plane.

Recall that for any complex number $u$, $|u|^2 = u\overline{u}$. Expanding the modulus squared of the sum and difference of $z$ and $w$:

1. $|z + w|^2 = (z + w)(\overline{z} + \overline{w}) = |z|^2 + |w|^2 + (z\overline{w} + \overline{z}w)$
2. $|z - w|^2 = (z - w)(\overline{z} - \overline{w}) = |z|^2 + |w|^2 - (z\overline{w} + \overline{z}w)$

Adding these two equations eliminates the cross terms $z\overline{w} + \overline{z}w$:

$$|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2)$$

Substitute the given values $|z| = 5$, $|w| = 7$, and $|z + w| = 10$:

$$10^2 + |z - w|^2 = 2\,(5^2 + 7^2)$$

$$100 + |z - w|^2 = 2(25 + 49) = 2(74) = 148$$

$$|z - w|^2 = 148 - 100 = 48$$

Taking the principal square root gives:

$$|z - w| = \sqrt{48} = 4\sqrt{3}$$

Common Misconceptions:

- Choice B ($2\sqrt{6}$) results from incorrectly computing $\sqrt{|w|^2 - |z|^2} = \sqrt{49 - 25} = \sqrt{24}$.
- Choice C ($4\sqrt{6}$) arises from mistakenly computing $\sqrt{10^2 - (7 - 5)^2} = \sqrt{96}$.
- Choice D ($2\sqrt{37}$) comes from calculating $\sqrt{2(|z|^2 + |w|^2)} = \sqrt{148}$ while forgetting to subtract $|z + w|^2$.

**7** MCQ

For the imaginary unit $i$, where $i^2 = -1$, which of the following complex numbers is equivalent to $(4 + 3i)(2 - 5i)$?

A $23 - 14i$ ✓

B $23 + 14i$

C $-7 - 14i$

D $-7 + 14i$

**WORKED SOLUTION & VERIFICATION**

To multiply the two complex numbers, apply the distributive property (FOIL):

$$(4 + 3i)(2 - 5i) = 4(2) + 4(-5i) + 3i(2) + 3i(-5i)$$

Compute each term:

$$= 8 - 20i + 6i - 15i^2$$

Since $i^2 = -1$, substitute $-15i^2 = -15(-1) = 15$:

$$= 8 - 14i + 15 = (8 + 15) - 14i = 23 - 14i$$

A common error is setting $i^2 = 1$, which incorrectly yields $8 - 15 - 14i = -7 - 14i$.

**8** MCQ

In the complex number system, let $i^2 = -1$. For what real value of $k$ is the expression

$$\frac{6+ki}{3+2i}$$

a purely imaginary number?

- **A** -9 ✓
- **B** -4
- **C** 4
- **D** 9

**WORKED SOLUTION & VERIFICATION**

To determine when the quotient is purely imaginary, multiply the numerator and denominator by the complex conjugate of the denominator, $3-2i$:

$$\frac{6+ki}{3+2i} = \frac{(6+ki)(3-2i)}{(3+2i)(3-2i)}$$

Expand the denominator using $i^2 = -1$:

$$(3+2i)(3-2i) = 3^2 - (2i)^2 = 9 - 4(-1) = 13$$

Expand the numerator:

$$(6+ki)(3-2i) = 18 - 12i + 3ki - 2ki^2 = 18 + 2k + (3k-12)i$$

Thus, the expression in standard form $a+bi$ is:

$$\frac{18+2k}{13} + \frac{3k-12}{13}i$$

A complex number is purely imaginary if and only if its real part is equal to $0$ and its imaginary part is nonzero:

$$\frac{18+2k}{13} = 0 \implies 18+2k = 0 \implies k = -9$$

Checking the imaginary part for $k = -9$:

$$\frac{3(-9)-12}{13} = \frac{-39}{13} = -3 \neq 0$$

So the expression simplifies to $-3i$, which is purely imaginary.
Common error: Setting the imaginary part to $0$ ($3k-12=0 \implies k=4$) makes the number purely real, not purely imaginary.

**9** MCQ

In the complex plane, let $z$ be a complex number that satisfies the equation

$$|z-4|+|z+4|=10.$$

What is the maximum possible value of $|z-3i|$?

A $\sqrt{34}$

B $\frac{13}{2}$

C $6$

D $\frac{25}{4}$ ✓

**WORKED SOLUTION & VERIFICATION**

1. **Identify the locus in the complex plane:**

The equation $|z-4|+|z+4|=10$ defines the set of all points $z=x+yi$ whose sum of distances to the fixed points $(4,0)$ and $(-4,0)$ is constant and equal to $10$. This is the definition of an ellipse with foci at $(\pm 4,0)$.

- Major axis length: $2a=10 \implies a=5$.
- Focal distance: $c=4$.
- Semi-minor axis: $b=\sqrt{a^2-c^2}=\sqrt{25-16}=3$.

Therefore, the equation of the ellipse in Cartesian coordinates is

$$\frac{x^2}{25}+\frac{y^2}{9}=1, \quad \text{where } -5\le x\le 5 \text{ and } -3\le y\le 3.$$

2. **Express the objective distance:**

We seek to maximize $|z-3i|$, which represents the Euclidean distance from $(x,y)$ on the ellipse to the point $(0,3)$:

$$|z-3i|^2=x^2+(y-3)^2.$$

From the ellipse equation, $x^2=25\left(1-\frac{y^2}{9}\right)=25-\frac{25}{9}y^2$.

3. **Set up the single-variable quadratic function:**

Substitute $x^2$ into the distance squared:

$$D^2(y)=\left(25-\frac{25}{9}y^2\right)+(y^2-6y+9)=34-\frac{16}{9}y^2-6y.$$

This is a downward-opening parabola in $y$. Its maximum occurs at the vertex:

$$y=-\frac{-6}{2\left(-\frac{16}{9}\right)}=-\frac{6}{\frac{32}{9}}=-\frac{27}{16}.$$

Notice that $-\frac{27}{16}=-1.6875$, which lies inside the valid interval $[-3,3]$.

4. **Evaluate the maximum distance:**

Substituting $y=-\frac{27}{16}$:

$$D^2\left(-\frac{27}{16}\right)=34-6\left(-\frac{27}{16}\right)-\frac{16}{9}\left(-\frac{27}{16}\right)^2=34+\frac{81}{8}-\frac{81}{16}=34+\frac{81}{16}=\frac{625}{16}.$$

Taking the square root gives:

$$D=\sqrt{\frac{625}{16}}=\frac{25}{4}=6.25.$$

5. **Analysis of common distractors:**

- $6$: Assumes the maximum must occur at the opposite vertical co-vertex $(0,-3)$, giving distance $|-3i-3i|=6$.
- $\sqrt{34}\approx 5.83$: Assumes the maximum occurs at the horizontal vertices $(\pm 5,0)$, where $|\pm 5-3i|=\sqrt{25+9}=\sqrt{34}$

.
Both are local boundary points smaller than the true maximum $\frac{25}{4} = 6.25$.

**10** MCQ

For the imaginary unit $i$, where $i^2 = -1$, which of the following is equivalent to the expression $(5 - 2i)(3 + 4i)$ ?

A $7 + 14i$

B $23 + 26i$

C $23 + 14i$ ✓

D $7 - 14i$

**WORKED SOLUTION & VERIFICATION**

To multiply the two complex numbers, apply the distributive property (FOIL method):

$$(5 - 2i)(3 + 4i) = (5)(3) + (5)(4i) + (-2i)(3) + (-2i)(4i)$$

Compute each term:
- First: $5 \times 3 = 15$
- Outside: $5 \times 4i = 20i$
- Inside: $-2i \times 3 = -6i$
- Last: $-2i \times 4i = -8i^2$

Substitute $i^2 = -1$:

$$-8i^2 = -8(-1) = 8$$

Now, group and combine the real and imaginary components:

$$\text{Real part: } 15 + 8 = 23$$

$$\text{Imaginary part: } 20i - 6i = 14i$$

Thus, the product is $23 + 14i$.
- Choice B ($7 + 14i$) arises from incorrectly assuming $i^2 = 1$, giving $15 - 8 = 7$.
- Choice C ($23 + 26i$) arises from failing to distribute the negative sign to $-2i \times 3$, giving $20i + 6i = 26i$.
- Choice D ($7 - 14i$) combines multiple sign errors.

**11** MCQ

What is $i^4$?

A $i$

B $-i$

C $-1$

D $1$ ✓

**WORKED SOLUTION & VERIFICATION**

Applying Complex Numbers principles, $i^4 = (i^2)^2 = (-1)^2 = 1$. Therefore, the verified correct option is $1$.

**12** MCQ

Simplify $(2+3i)+(5-i)$.

- **A** $7+2i$ ✓
- **B** $-3+2i$
- **C** $7-2i$
- **D** $7+4i$

**WORKED SOLUTION & VERIFICATION**

Add real and imaginary parts separately: $(2+5)+(3-1)i=7+2i$.

**13** MCQ

Simplify $(4-2i)-(1+3i)$.

- **A** $3+i$
- **B** $5-5i$
- **C** $3-5i$ ✓
- **D** $3+5i$

**WORKED SOLUTION & VERIFICATION**

Applying Complex Numbers principles, $(4-1)+(-2-3)i=3-5i$. Therefore, the verified correct option is $3-5i$.

**14** MCQ

What is the imaginary part of $z=6-7i$?

- **A** $6$
- **B** $-7$ ✓
- **C** $7$
- **D** $-6$

**WORKED SOLUTION & VERIFICATION**

The imaginary part is the real coefficient of $i$, which is $-7$.

**15** MCQ

Simplify $i\cdot i\cdot i$.

- **A** $i$
- **B** $1$
- **C** $-i$ ✓
- **D** $-1$

**WORKED SOLUTION & VERIFICATION**

Applying Complex Numbers principles, $i^3=i^2\cdot i=(-1)(i)=-i$. Therefore, the verified correct option is $-i$.

**16** MCQ

Simplify $3i\cdot 4i$.

- **A** $12$
- **B** $12i$
- **C** $-12$ ✓
- **D** $-12i$

**WORKED SOLUTION & VERIFICATION**

Applying Complex Numbers principles, $3i\cdot 4i=12i^2=12(-1)=-12$. Therefore, the verified correct option is $-12$.

**17** MCQ

Find $|3+4i|$.

- **A** $5$ ✓
- **B** $7$
- **C** $25$
- **D** $\sqrt{7}$

**WORKED SOLUTION & VERIFICATION**

$|3+4i| = \sqrt{3^2+4^2} = \sqrt{25} = 5.$

**18** MCQ

What is the complex conjugate of $5-2i$?

- **A** $5-2i$
- **B** $-5-2i$
- **C** $5+2i$ ✓
- **D** $-5+2i$

**WORKED SOLUTION & VERIFICATION**

The conjugate flips the sign of the imaginary part only: $5-2i \to 5+2i$.

**19** MCQ

Multiply $(2+i)(3-i)$.

- **A** $7-i$
- **B** $5+i$
- **C** $7+i$ ✓
- **D** $6+i$

**WORKED SOLUTION & VERIFICATION**

$(2+i)(3-i) = 6-2i+3i-i^2 = 6+i+1 = 7+i.$

**20** MCQ

Simplify $\dfrac{1+i}{1-i}$.

- **A** $-i$
- **B** $1$
- **C** $i$ ✓
- **D** $-1$

**WORKED SOLUTION & VERIFICATION**

Multiply by the conjugate: $\dfrac{(1+i)^2}{(1-i)(1+i)} = \dfrac{2i}{2} = i.$

**21** MCQ

Simplify $i^7$.

A $i$

B $-i$ ✓

C $1$

D $-1$

**WORKED SOLUTION & VERIFICATION**

Powers of $i$ cycle every 4: $i^7 = i^4 \cdot i^3 = 1 \cdot (-i) = -i$.

**22** MCQ

Find $|z|$ for $z = 5 - 12i$.

A $\sqrt{17}$

B $17$

C $7$

D $13$ ✓

**WORKED SOLUTION & VERIFICATION**

$|5 - 12i| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

**23** MCQ

If $(x + 2) + (y - 3)i = 5 + 4i$ for real $x, y$, find $x + y$.

A $9$

B $10$ ✓

C $8$

D $12$

**WORKED SOLUTION & VERIFICATION**

Equate real and imaginary parts: $x + 2 = 5 \Rightarrow x = 3$; $y - 3 = 4 \Rightarrow y = 7$. So $x + y = 10$.

**24** MCQ

Simplify $(2 - 3i)^2$.

A $-5 + 12i$

B $13 - 12i$

C $-5 - 12i$ ✓

D $5 - 12i$

**WORKED SOLUTION & VERIFICATION**

$(2 - 3i)^2 = 4 - 12i + 9i^2 = 4 - 12i - 9 = -5 - 12i$.

**25** MCQ

If $z = 4 + 3i$, find $z\bar{z}$.

A $5$

B $-7$

C $25$ ✓

D $7$

**WORKED SOLUTION & VERIFICATION**

Applying Complex Numbers principles, $z\bar{z} = |z|^2 = 4^2 + 3^2 = 16 + 9 = 25$. Therefore, the verified correct option is 25.